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The Complete Guide

ALEXANDER DÜRING



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Fixed Income Trading and Risk Management

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ALEXANDER DÜRING

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Foreword



Source: ALEXANDER DÜRING

This book about fixed income has been inspired by over two decades of work in the market from offices in London, Frankfurt and Tokyo. During this time, the author had the good fortune to be close to, and sometimes involved in, an eclectic mix of events. Although working for a large investment bank and the European Central Bank has made it possible to collect this experience, the book does not represent the views of either. Any opinions about central banking should in particular not be taken as those of the Eurosystem.

This book is also inspired by a building, namely the British Museum in London, a detail of which is pictured above. This inspiration has three aspects. The first is that the architecture of the building cannot be understood by looking only at its historic base, or only at Sir Norman Foster's modern additions. Fixed income markets can similarly not be properly understood by considering only their conventions, or only the modern mathematical apparatus that has been created to analyse them. Although this book will present complex mathematics determining the risk characteristics of various instruments, some design features of these instruments are archaic. These features remain today for no better reason than that there was never a need to change them.

Second, the museum is open to all and free in principle, although a small donation is encouraged. Children can walk in and experience artefacts that once were created for the exclusive use of priests and kings. In the world of fixed income, the efforts of statistics offices, central banks, trade associations and others have made a wealth of data available to the general public. Freely available numerical libraries and advances in computers make it possible for everyone to analyse, and learn from that data. Understanding of markets has thus been democratised in ways that were unimaginable even a decade ago. This book, written at home using only freely available and self-developed software, is built on these foundations and encourages such exploration.

Third, the British Museum is not uncontroversial, as are some of the exhibits shown there. The ideas of debt, of trading debt, and of banking are similarly the subject of debate and argument. This book gives some space to this debate although the author would not claim to be neutral in it. The financial industry is part of the service sector, however else it may be portrayed in popular culture. People working in it cannot be oblivious to their role in society, and take it as granted that their actions will not be questioned by others.

Books not only require inspiration, but also enablement. In this case, this has been provided by my family who kindly tolerated the amount of time I spent on writing it and whom I thank for their forbearance. Erwin, Reiko and George at Deutsche, and Julian at the ECB helped me to develop a somewhat non-conformist approach to financial markets. Tamio and Henry at Deutsche Securities Inc. encouraged me to teach, as did Toto, Christophe and Ralph at the ECB. Those who attended these classes and asked questions will hopefully find them answered here. To them, and many not named, go my thanks for making work in financial markets the joy it is.

As this is a textbook, literature references are given here extensively with the aim of encouraging further reading rather than striving for completeness in representing the state of the art.